

# Physics-Informed Neural Networks for Fluid Flow Prediction

## Project Notes and Technical Report

### Introduction

This project investigates the application of Physics-Informed Neural Networks (PINNs) for fluid flow prediction.

The primary objective is to compare a standard Neural Network (NN) with a Physics-Informed Neural Network and evaluate whether incorporating physical laws into the training process improves prediction accuracy when only a limited amount of training data is available.

Fluid dynamics plays a critical role in many engineering and scientific applications, including aerospace engineering, weather forecasting, biomedical engineering, and industrial process design.

Traditional machine learning models learn patterns solely from data. However, physical systems are governed by mathematical laws. Physics-Informed Neural Networks combine both sources of information by learning from observational data while simultaneously satisfying governing physical equations.

The central research question addressed in this study is:

*Can a Physics-Informed Neural Network predict fluid flow more accurately than a standard Neural Network when training data is scarce?*

### Motivation

One of the major limitations of conventional machine learning models is their dependence on large amounts of labeled training data.

In many engineering applications, acquiring data can be expensive because experiments, measurements, and high-fidelity simulations require significant computational and financial resources.

At the same time, the governing physical laws of the system are often known.

For fluid flow problems, conservation principles such as conservation of mass and momentum describe the underlying dynamics.

Rather than ignoring this information, Physics-Informed Neural Networks integrate these laws directly into the learning process.

As a result, PINNs can achieve improved generalization and physically consistent predictions even when only a small amount of data is available.

## Dataset Description

The dataset used in this study was generated using Computational Fluid Dynamics (CFD) simulations.

Each sample contains spatial coordinates and time information describing the state of the fluid.

The input variables are:

- $x$  coordinate
- $y$  coordinate
- Time  $t$

The output variables are:

- Horizontal velocity component  $u$
- Vertical velocity component  $v$

The dataset also contains spatial derivatives:

$$\frac{\partial u}{\partial x}, \quad \frac{\partial u}{\partial y}, \quad \frac{\partial v}{\partial x}, \quad \frac{\partial v}{\partial y}$$

These derivatives describe how the velocity field changes throughout the domain.

The complete dataset contains approximately 10,000 samples.

The availability of derivative information makes the dataset particularly suitable for training Physics-Informed Neural Networks.

## Project Resources

### Google Colab Notebook

The complete implementation can be accessed through the following Google Colab notebook:

[https://colab.research.google.com/drive/1KEBy-XeMXGZFZ5GQJMMofitAJ\\_Aa2iUc?usp=sharing](https://colab.research.google.com/drive/1KEBy-XeMXGZFZ5GQJMMofitAJ_Aa2iUc?usp=sharing)

The notebook contains:

- Data preprocessing
- Standard Neural Network implementation
- Physics-Informed Neural Network implementation
- Automatic differentiation using PyTorch
- Training and evaluation routines
- Visualization and plotting code

### Dataset Source

The CFD dataset used in this study was obtained from Kaggle:

<https://www.kaggle.com/datasets/allanwandia/computational-fluid-dynamics>

The dataset contains:

- Spatial coordinates  $(x, y)$
- Time values  $(t)$
- Velocity components  $(u, v)$
- Velocity derivative information

## Problem Statement

The objective is to learn the mapping

$$(x, y, t) \rightarrow (u, v)$$

In other words, given a spatial location and time, the model should predict the corresponding fluid velocity components.

Two models were developed and compared:

1. Standard Neural Network (NN)
2. Physics-Informed Neural Network (PINN)

Performance was evaluated using Root Mean Square Error (RMSE).

## Standard Neural Network

The first model is a fully connected feedforward neural network.

Architecture:

- Input layer: 3 neurons
- Hidden layers: 3
- Neurons per hidden layer: 64
- Activation function: Tanh
- Output layer: 2 neurons

Inputs:

$$(x, y, t)$$

Outputs:

$$(\hat{u}, \hat{v})$$

The model is trained using Mean Squared Error (MSE) loss:

$$L_{data} = \frac{1}{N} \sum_{i=1}^N [(u_i - \hat{u}_i)^2 + (v_i - \hat{v}_i)^2]$$

The neural network learns exclusively from data and has no knowledge of fluid mechanics.

## Physics-Informed Neural Network

The PINN uses the same neural network architecture.

The difference lies in the inclusion of physics-based constraints during training.

The total loss function is

$$L_{total} = L_{data} + \lambda L_{physics}$$

where

- $L_{data}$  measures prediction accuracy
- $L_{physics}$  measures compliance with physical laws
- $\lambda$  controls the influence of physics constraints

The physics loss acts as a regularizer and guides the model toward physically meaningful solutions.

## Physics Constraints

The primary physical constraint used in this work is the incompressible continuity equation:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

This equation represents conservation of mass.

To enforce this condition, the continuity residual is defined as

$$R_c = \frac{\partial \hat{u}}{\partial x} + \frac{\partial \hat{v}}{\partial y}$$

The residual should ideally be zero throughout the domain.

The dataset also provides velocity derivatives.

Using automatic differentiation, the PINN computes

$$\frac{\partial \hat{u}}{\partial x}, \quad \frac{\partial \hat{u}}{\partial y}, \quad \frac{\partial \hat{v}}{\partial x}, \quad \frac{\partial \hat{v}}{\partial y}$$

These predicted derivatives are compared with the ground-truth derivatives available in the dataset.

Derivative matching encourages the model to learn realistic fluid dynamics instead of merely fitting velocity values.

## Automatic Differentiation

A key feature of PINNs is automatic differentiation.

PyTorch computes derivatives directly through the computational graph rather than approximating them numerically.

Examples include:

$$\frac{\partial \hat{u}}{\partial x}$$

and

$$\frac{\partial \hat{v}}{\partial y}$$

These derivatives are required to evaluate the continuity equation and other physical constraints during training.

Automatic differentiation enables efficient implementation of physics-based loss functions.

## Training Strategy

To investigate the effectiveness of PINNs under limited-data conditions, only 1% of the dataset was used for training.

This corresponds to approximately 100 samples out of 10,000.

The remaining samples were reserved for testing and evaluation.

This sparse-data scenario was intentionally selected because standard neural networks often struggle when training data is limited.

PINNs are expected to perform better because physical laws provide additional information during optimization.

## Results and Discussion

Experiments were performed using different values of the physics-loss weighting parameter  $\lambda$ .

Performance was measured using RMSE for both velocity components.

| $\lambda$ | NN RMSE U | NN RMSE V | PINN RMSE U | PINN RMSE V |
|-----------|-----------|-----------|-------------|-------------|
| 0.001     | 0.1913    | 0.1699    | 0.1917      | 0.1355      |
| 0.01      | 0.2078    | 0.1619    | 0.2424      | 0.1340      |
| 0.1       | 0.2015    | 0.1629    | 0.1857      | 0.1065      |
| 1         | 0.1834    | 0.1557    | 0.2415      | 0.0980      |
| 10        | 0.1929    | 0.1591    | 0.5525      | 0.0979      |

### Effect on U Velocity Component

The best PINN performance for the  $u$  velocity component was achieved at

$$\lambda = 0.1$$

with

$$RMSE_u = 0.1857$$

compared with the Neural Network error

$$RMSE_u = 0.2015$$

at the same value of  $\lambda$ .

When the physics loss becomes excessively dominant, prediction accuracy deteriorates.

For example,

$$\lambda = 10$$

produced

$$RMSE_u = 0.5525$$

indicating over-constrained learning.

### **Effect on V Velocity Component**

For the  $v$  velocity component, the PINN consistently outperformed the standard Neural Network.

The best performance was achieved at

$$\lambda = 1$$

with

$$RMSE_v = 0.0980$$

compared to

$$RMSE_v = 0.1557$$

for the standard Neural Network.

This represents a significant reduction in prediction error.

## Overall Analysis

Several observations can be made:

1. PINNs consistently improved prediction accuracy for the  $v$  velocity component.
2. Moderate values of  $\lambda$  produced the best balance between data fitting and physical consistency.
3. Very large values of  $\lambda$  can over-constrain the model and reduce predictive accuracy.
4. The best overall trade-off was observed around

$$\lambda = 0.1$$

where both accuracy and physical consistency were maintained.

These results demonstrate the importance of carefully selecting the physics-loss weighting parameter.

## Visualizations

Several visualization techniques were used to evaluate model performance:

- Velocity vector field plots
- Velocity magnitude contour plots
- Streamline plots
- RMSE comparison charts
- Error heatmaps

These visualizations provide insight into the physical realism of the predicted flow fields.

In most cases, the PINN produced smoother and more physically consistent flow structures.

## Advantages of PINNs

Physics-Informed Neural Networks provide several advantages:

- Effective learning with limited data
- Incorporation of scientific knowledge
- Improved physical consistency
- Better generalization capability
- Ability to estimate difficult-to-measure quantities

These properties make PINNs attractive for scientific machine learning and engineering applications.

## Conclusion

This study compared a standard Neural Network and a Physics-Informed Neural Network for predicting fluid velocity fields from sparse Computational Fluid Dynamics data.

The Physics-Informed Neural Network incorporated physical knowledge through the incompressible continuity equation and derivative matching constraints. By combining observational data with governing physical laws, the PINN was able to learn more physically consistent solutions.

Experimental results showed that PINNs significantly improved prediction accuracy for the  $v$  velocity component and achieved the best overall balance between accuracy and physical consistency when the physics-loss weighting parameter was set to

$$\lambda = 0.1.$$

The findings highlight the importance of integrating scientific knowledge into machine learning models, particularly in situations where training data is limited.

Overall, Physics-Informed Neural Networks represent a promising approach for scientific machine learning, fluid dynamics, and engineering applications where physical laws are known but large datasets are unavailable.